

Business Intelligence Project

Problem Statement

A healthcare organization requires a centralized reporting system to track drug products, manufacturers, approval status, and market availability. Due to poor data quality and inconsistent records, stakeholders are unable to generate reliable business insights.

The organization seeks a data analyst to transform raw pharmaceutical data into a structured dataset and develop dashboards that answer key business questions such as:

- Which manufacturers have the largest product portfolios?
- How have FDA approvals changed over time?
- Which drug categories dominate the market?
- What percentage of products remain active versus terminated?
- What trends exist in product purchases and coverage?

Project Deliverables

1. Data Quality Assessment Report

A detailed report identifying:

- Missing values
- Duplicate records
- Inconsistent formats
- Invalid date values
- Empty or redundant columns
- Data quality recommendations

Deliverable: Data Quality Assessment Document

2. Cleaned and Structured Dataset

Prepare an analysis-ready dataset by:

- Removing duplicates
- Handling missing values
- Standardizing text fields
- Formatting dates
- Creating derived columns
- Validating records

Deliverable: Cleaned Excel/CSV file

3. Data Dictionary

Documentation explaining:

Column Name	Description	Data Type
FDA Name	Drug Name	Text
Labeler Name	Manufacturer	Text
FDA Approval Date	Approval Date	Date

Deliverable: Data Dictionary Document

4. Exploratory Data Analysis (EDA)

Perform analysis to answer:

- Which manufacturers have the most products?
- Which drug categories are most common?
- How have approvals changed over time?
- What percentage of drugs are active?
- What trends exist in purchases and coverage?

Deliverable: EDA Report with findings and visualizations

5. KPI Development

Create key performance indicators:

- Total Drug Products
- Active Drugs
- Terminated Drugs
- Total Manufacturers
- FDA Approvals by Year
- Purchased Products
- Coverage Percentage

Deliverable: KPI Summary Sheet

6. MYSQL

- Create MySQL Database
 - Create Table (based on your project)
 - Import Excel File
 - Connect MySQL to Power BI
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7. Interactive Dashboard

Develop a dashboard using Excel, Power BI, or Tableau containing:

Executive KPIs

- Total Drugs
- Active Drugs
- Manufacturers
- Approval Count

Visualizations

- Approval Trend Analysis
- Manufacturer Ranking
- Drug Category Distribution
- Therapeutic Equivalence Analysis
- Purchase Trends
- Coverage Analysis

Deliverable: Interactive Dashboard

8. Business Insights Report

Provide insights and recommendations such as:

- Top-performing manufacturers
- Growth in FDA approvals
- Underrepresented drug categories
- Coverage gaps
- Product lifecycle observations

Deliverable: Business Insights & Recommendations Report

9. Final Project Presentation

A professional presentation including:

- Business Problem
- Dataset Overview
- Data Cleaning Process
- Analysis Findings
- Dashboard Walkthrough
- Recommendations

Deliverable: PowerPoint Presentation (10–15 slides)

